Sales and Profit Analysis for Neo coffee shop

Problem statement

Running a coffee shop is not just about making good coffee. Daily sales can change a lot depending on customer behavior, product choice, and time of year. While coffee shops collect sales data from transactions, this data is often not fully used to understand what is actually driving revenue and profit.

This project focuses on using past sales data to understand how the coffee shop is performing over time. The goal is to identify patterns in sales and profit, see which products are most profitable, and understand how customer behavior differs across segments such as loyalty card holders and non-loyalty customers.

The main question this analysis aims to answer is: How do sales and profit change over time in Neo's coffee shop, and which products and customer groups contribute most to overall performance?

Analytical Objectives

The objectives of this analysis are:

• To examine how sales, revenue, and profit change over time using historical transaction data

• To identify which coffee types, roast types, and sizes generate the most profit

• To compare purchasing behavior between loyalty card holders and customers without loyalty cards

• To check whether observed differences in profit and sales are statistically meaningful

• To present insights that can help improve pricing, product focus, and customer retention strategies

Description of the Data Source

This analysis will use a coffee shop transactional dataset stored in a CSV file (Neo_Coffee.csv) obtained from github.

Each row in the dataset represents a single product purchased in an order.
The dataset contains transaction records from 2021 to 2023, based on the order date. It includes information on sales, customers, and products.
Key variables in the dataset include:

• Order information: order ID and order date

- Sales and pricing: quantity sold, unit price, price per 100g, and profit
- Product details: coffee type, roast type, and size
- Customer information: customer ID, loyalty card status, city, and country

Proposed Metrics and Statistical Methods

Key Metrics.
The following metrics will be used in the analysis:

- Total revenue and total profit over time
- Number of orders and total quantity sold
- Average order value
- Profit contribution by product type and coffee size
- Comparison of revenue and profit between loyalty and non-loyalty customers.

Time-based metrics such as monthly growth and rolling averages will be used to better understand trends and smooth short-term fluctuations.

Statistical Methods.
To support the analysis, the following methods will be applied:

- Descriptive statistics to summarize sales, profit, and customer behavior
- Independent t-tests to compare average profit between loyalty and non-loyalty customers
- Analysis of variance (ANOVA) to compare profit across different coffee types, roast types, and sizes
- Trend analysis and visualization to explore changes in sales and profit over time

These methods will help determine whether observed differences are meaningful rather than due to chance.

Risks and Constraints

There are several limitations to this analysis:

• The dataset does not include operating costs such as rent, staff wages, or utilities, so profit is analyzed only based on the available data

• Marketing activities, promotions, and external factors such as holidays or weather are not captured

• The data represents transactions from a single coffee shop, which limits how widely the results can be generalized

• The analysis focuses on patterns and relationships and does not establish cause-and-effect conclusions

Despite these limitations, the dataset is suitable for understanding sales performance and customer behavior over time.